newer terms and files needed for editing.

> **Note:** The main configuration file grub.cfg is automatically generated; thus the changes made are temporary and tend to fade away with *grub update*. During Grub2 configuration the most important thing is to avoid making changes directly to the grub.cfg file.

Adding custom entries: Even though the detection mechanism is robust and efficient, there may be a scenario when Grub fails to add an entry or you want to add your own entry. For adding a custom entry in Grub2, you have to modify the file */etc/grub.d/40_custom*. The format in which the entry should be written is as follows:

- menuentry "name": This is the name that is shown in the Grub2 bootloader for selection.
- set root=(hdx, y): Defines the partition from where the boot files are loaded and booting has progressed, where X denotes the disk number and Y the partition number.
- linux /vmlinuz-xyz: Defines the path to the Linux image; add this if you are adding a Linux entry.
- initrd /initramfs.img: Defines the *ramfs* for the system, which is only needed if the system depends on *initrd* for booting. It is not required for Windows systems.

Here is what our */etc/grub.d/40_custom* looks like after making a user defined Grub entry:

```
# (1) OSFY_Linux
menuentry "Open Source For You" {
set root=(hd0,2)
linux /vmlinuz-linux root=/dev/sda2 ro nomodeset
initrd /initramfs-linux26.img
}
```

To append these changes in the main configuration file, run the *grub update* commands. Grub2 can be updated by using two commands. As root user, issue any of the following commands:

```
update-grub
mkconfig-grub –o /boot/grub/grub.cfg
```

Booting an ISO with Grub2: With the Grub2 custom menu and entry wizard, booting an ISO directly from Grub2 has become much easier. There is no need to burn an optical disk when you have a nifty bootloader at your disposal. Booting an ISO is very much like adding an entry, the only difference is that rather than pointing to a kernel, you point to the ISO for booting. To boot an Ubuntu ISO, append the following statement in the */etc/grub.d/40_custom* file:

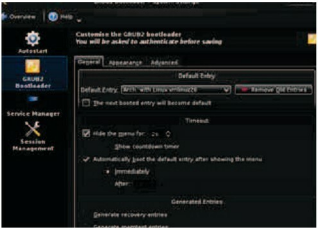

Figure 2: Grub2 bootloader editor

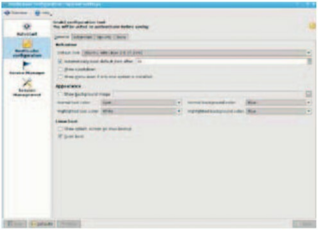

Figure 3: Grub2 KCm

```
menuentry "ubuntu-11.04-desktop-amd64.iso" {
set isofile="/media/hd1/ubuntu-11.04-desktop-amd64.iso"
loopback loop (hd0,1)$isofile
linux (loop)/casper/vmlinuz boot=casper iso-scan/
filename=$isofile quiet noeject noprompt splash --
initrd (loop)/casper/initrd.lz
}
```

Change the path of the ISO and replace the (hd0,1) where 0 is the disk number and 1 is the partition number.

Securing Grub2: Security was one of the key features underlying Grub2 development. Grub2 boasts of a highly secure system. To further aid security, you can define a password and allow users to boot into a specified OS. Grub2 is still in the active development stage, so settings can vary depending on how developers want to streamline things.

Grub2 offers multiple solutions for displaying Grub entries. A user can have both protected and unprotected entries in the Grub2 boot loader menu.

The Grub2 securing system takes a step forward by allowing you to select users and assign them different booting privileges. From a secure point of view Grub2 offers multiple users and Grub configuration files, making it more perplexing for users to tinker with the settings on their own.